Practical Session 5 - NP-Completeness

Dr. Simone Fioravanti
07/05/2026

Problem 1

Consider the following decision problem. Clique: Given an undirected graph $G = (V, E)$ and an integer k, does there exist a subset of vertices $S \subseteq V$ of size $|S| \geq k$ that induces a complete subgraph i.e. a *clique*?
Prove that Clique is NP-complete via a reduction from 3-SAT.

Problem 2

Prove that Clique $\leq_p$ IndependentSet.

Problem 3

Consider the following decision problem. 4D-Matching: Given X, Y, Z, W disjoint sets of cardinality n and a set of quadruples $Q \subseteq X \times Y \times Z \times W$, is there a subset $Q' \subseteq Q$ such that each element of $X \cup Y \cup Z \cup W$ appears exactly in one tuple in Q'?
Prove that 4D-Matching is NP-complete.

Problem 4 (Bonus)

Suppose you're helping to organize a summer sports camp, and the following problem comes up. The camp is supposed to have at least one counselor who's skilled at each of the n sports covered by the camp (baseball, volleyball, and so on). They have received job applications from m potential counselors. For each of the n sports, there is some subset of the m applicants qualified in that sport. The question is: For a given number $k < m$, is it possible to hire at most k of the counselors and have at least one counselor qualified in each of the n sports? We'll call this the EfficientRecruiting Problem. Show that EfficientRecruiting is NP-complete.

Solution: EfficientRecruiting is clearly in NP. It is polynomial to check if a proposed set of counselors are qualified for all the sports.

To prove NP-hardness, we show a reduction from VertexCover (with parameter k) to EfficientRecruiting with the same parameter. From a graph $G = (V, E)$ we build the corresponding instance of EfficientRecruiting in this way: each vertex $v \in V$ is a counselor and each $e \in E$ is a sport. So we have exactly 2 counselors for each sport.

A vertex cover of size k exists if and only if a set of counselors of size k exists as well. If we have a vertex cover $V' \subseteq V$ of size k it means that every edge is covered by a $v \in V'$. If we take the counselors corresponding to the vertices in V' they will be qualified in all sports. The opposite direction is similar.

Problem 5 (Bonus)

Consider the following decision problem. Max2SAT: Given a CNF formula consisting of clauses $C = C_1, C_2, \ldots, C_k$ and literals $X = x_1, x_2, \ldots, x_n$ such that each clause involves exactly two literals, does there exist an assignment of TRUE/FALSE to the literals such that at least k' clauses are satisfied?

Prove that Max2SAT is NP-complete via a reduction from Clique.

Solution: In the following, I will show the reduction. Proving that it is in NP is very easy and left to the reader. An instance of Clique is determined by a graph $G = (V, E)$ and a number k. Let us call $\bar{E} = V \times V \setminus E$. We build an instance of Max2SAT, in the following way:

(i) For each non-edge $(v_i, v_j) \in \bar{E}$ we add a clause $C_{ij} = \bar{x}_i \vee \bar{x}_j$.

(ii) For each $v_i \in V$ we add two clauses $(x_i \vee z) \wedge (x_i \vee \bar{z})$ where z is a dummy variable.

Also, we define $k' = |V| + k + |\bar{E}|$. We will show that a clique of size $\geq k$ exists if and only if at least k' clauses are satisfied.

($\Rightarrow$): Let $W = \{w_1, \dots, w_k\}$ be a clique of size $k^* \geq k$. We set all variables corresponding to those nodes to 1 and the others to 0 and z to 1 as well. Let us call them $x_1, \dots, x_{k^*}$.

(i) Each non-edge clause is satisfied. Indeed, since W is a clique, at least one of the 2 literals in such a clause is set to 0 (because it corresponds to a vertex that is not in the clique), thus its negation is 1 and the clause is satisfied. In total, they are $|\bar{E}|$

(ii) The clauses of these kind $(x_i \vee z) \wedge (x_i \vee \bar{z})$ will be both satisfied if $i = 1, \dots, k^*$, while, otherwise, only one of them will be (i.e. the first). In total, they are $2k^* + (|V| - k^*) = k^* + |V|$.

We have obtained $k^* + |V| + |\bar{E}| \geq k'$ satisfied clauses.

($\Leftarrow$): Now, let vector $\mathbf{x} \in \{0, 1\}^{|V|}$ be a truth assignment for the variables x_i such that the number of clauses satisfied is at least k'. Since the clauses containing z are symmetrical, let us fix $z = 1$. We define the candidate vertex set $W(\mathbf{x})$ as the subset of vertices whose corresponding variables are set to 1:

$$W(\mathbf{x}) = \{v_i \in V \mid x_i = 1\}$$

By construction, **if all non-edge clauses** are satisfied, then for every pair $v_i, v_j \in W(\mathbf{x})$, the edge (v_i, v_j) must exist in E. Otherwise, the clause $(\bar{x}_i \vee \bar{x}_j)$ would be evaluated as 0 (or FALSE). Thus, $W(\mathbf{x})$ induces a clique in G.

However, in principle this could not correspond to a clique: if $W(\mathbf{x})$ for instance, is composed of a clique of size k plus a node v such that v is connected to **all** nodes in the clique, except one w, we would satisfy $\geq k'$ clauses as well: the clause corresponding to the non-edge (v, w) will not be satisfied but the clauses with z will be both satisfied, counterbalancing the number.

Our aim is to transform $\mathbf{x}$ into a $\mathbf{x}^*$ such that $W(\mathbf{x}^*)$ is a clique. For each $\bar{x}_i \vee \bar{x}_j$ that is not satisfied, we arbitrarily set x_i to 0. In this way, we augment by 1 the number of non-edge clauses satisfied, and diminish by 1 the number of other clauses satisfied. In principle the number of non-edge clauses satisfied can increment by more than 1. In all cases, after the transformation the number of satisfied clauses in total stays $\geq k'$. We can do that for a polynomial number of times until all non-edge clauses are satisfied. If we call this assignment $\mathbf{x}^*$, the corresponding $W(\mathbf{x}^*)$ is a clique of size at least k.